

Javascript

BOM object

Theory

1. Window Object

- **Global object** in browsers.
- Common methods:
 - `window.open("url")` → opens a new tab/window.
 - `window.close()` → closes current window.
 - `window.alert("msg")` → shows alert box.
- Properties: `innerHeight`, `innerWidth`.

2. Screen Object

- Provides info about user's screen.
- Properties:
 - `screen.height`, `screen.width`.
 - `screen.availHeight`, `screen.availWidth`.

3. Location Object

- Represents current URL.
- Properties:
 - `location.href`, `location.host`, `location.pathname`, `location.protocol`.
- Methods:
 - `location.assign("url")` → navigates to new page.
 - `location.reload()` → reloads current page.

4. Timers

- `setTimeout(fn, ms)` → runs once after delay.
- `setInterval(fn, ms)` → runs repeatedly.
- `clearInterval(id)` → stops interval.

5. Storage

- **localStorage** → persists data even after browser closes.
- **sessionStorage** → persists data only until tab is closed.
- Methods: `setItem()`, `getItem()`, `removeItem()`, `clear()`.

6. Cookies

- Stored as `document.cookie = "key=value"`.
- Used for session management, personalization, tracking.



Coding Round Questions

- Window object → What are common properties and methods of the window object?
- Screen object → How do you get screen dimensions in JavaScript?
- Location object → What is the difference between `location.href` and `location.assign()`?
- Timers → Explain difference between `setTimeout` and `setInterval`.
- Storage → Compare `localStorage` and `sessionStorage`.
- Cookies → How do you set and read cookies in JavaScript?



Machine Coding Round Challenge

Problem: User Session Tracker

Requirements:

1. On page load, store user info in `localStorage`.
2. Track how many times the user has visited the page.
3. Display visit count dynamically.
4. Provide a button to clear session data.
5. Use cookies to store a username.

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>BOM</title>
  </head>
  <body>
    <h2>Browser Object Model</h2>
    <!-- <a href="https://www.google.com">Google</a> -->

    <button id="button">Go to google</button>

    <script>
      // window.alert("Hello welcome to our page");

      document.getElementById("button").onclick = () => {
        window.open("https://www.google.com");
      };

      // window.close();

      console.log(window.innerHeight);
      console.log(window.innerWidth);

      console.log("Screen height", screen.height);

      console.log("screen width", screen.width);

      console.log("availheight", screen.availHeight);

      console.log("availwidth", screen.availWidth);

      console.log(location.href);

      console.log(location.host);

      console.log(location.pathname);
      console.log(location.protocol);

      // location.assign("https://www.google.com")

      // location.reload();

      setTimeout(() => {
        console.log("Settimeout");
      }, 2000);
```

```
const id = setInterval(() => {
  console.log("Setinterval");
}, 1000);

setTimeout(() => {
  clearInterval(id);
}, 10000);

// localStorage.setItem("name", "Brindha");

// const result = localStorage.getItem("name");
// console.log(result);

// localStorage.removeItem("name");

// localStorage.clear();

let user = {
  name: "Brindha",
  age: 22,
  degree: "Bsc Maths",
};

localStorage.setItem("users", JSON.stringify(user));

const result = JSON.parse(localStorage.getItem("users"));

console.log(result);

// sessionStorage.setItem("Sessionuser", "Brindha");
sessionStorage.removeItem("Sessionuser");

sessionStorage.clear();

document.cookie = "username=Brindha";

console.log(document.cookie);

</script>
</body>
</html>
```